

CS 401

CS 401: Tagged PDF & Accessibility Notes

Demonstration of LaTeX Tagged PDF Accessibility Compliance

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Contents

LECTURE 1

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LaTeX Tagged PDF & PDF/UA Accessibility

This document demonstrates full compatibility with the official $\text{\LaTeX}$ Tagged PDF project (`\DocumentMetadata{tagging=on}`).

When tagged PDF mode is active, paragraph tagging around verbatim blocks (`\lstlisting`) and custom boxes (`codebox`) can be temporarily suspended using `\tagpdfsetup{para/tagging=false}`.

[DEF] Definition 1 (Tagged PDF)

A **Tagged PDF** is a PDF document that contains an underlying logical structure tree describing the roles and nesting of document components (such as headings, paragraphs, lists, and code snippets) for assistive technologies.

[THM] Theorem 1 (Accessibility Compliance)

All core environments in `modernclassnotes`—including callout boxes, theorems, exercises, solutions, and code boxes—are structurally compatible with $\text{\LaTeX}$'s `tagpdf` engine.

Proof: Internal paragraph sockets and verbatim line-break boxes in `modernclassnotes` automatically handle `tagpdf` hook state transitions, ensuring balanced structural trees during PDF compilation. ■

Python Implementation: Accessibility Metadata Validation

```
1 def check_pdf_accessibility(document):
2     """
3     Validates PDF structural tags for accessibility compliance.
4     """
5     has_tagged_tree = document.has_struct_tree_root()
6     has_language_meta = document.get_language() is not None
7
8     return has_tagged_tree and has_language_meta
```

[*] Pro Tip

To build accessible lecture notes for students using screen readers, simply place `\DocumentMetadata{tagging=on}` at the very top of your document before `\documentclass`.

[KEY] Key Takeaways

- `\DocumentMetadata{tagging=on}` enables PDF/UA accessibility metadata.
- `modernclassnotes` supports both standard and tagged compilation modes natively.
- Visual styling remains completely identical across both modes.